

**Batch: C3      Roll No.: 16010123325**

**Experiment No.**

**Grade: AA / AB / BB / BC / CC / CD / DD**

**Signature of the Staff In-charge with date**

**Title:** Write a program to Compute convolution correlation of two discrete time signal sequences using Matlab.

**Objective:** To familiarize the beginner to MATLAB by introducing the basic features and commands of the program.

**Expected Outcome of Experiment:**

| CO  | Outcome                                                                                                           |
|-----|-------------------------------------------------------------------------------------------------------------------|
| CO3 | To understand the concept of convolution and perform different convolution operations on the given input signals. |

**Books/ Journals/ Websites referred:**

1. <http://www.mathworks.com/support/>
2. [www.math.mtu.edu/~msgocken/intro/intro.html](http://www.math.mtu.edu/~msgocken/intro/intro.html)
3. [www.mccormick.northwestern.edu/docs/efirst/matlab.pdf](http://www.mccormick.northwestern.edu/docs/efirst/matlab.pdf)
4. A.Nagoor Kani “Digital Signal Processing”, 2<sup>nd</sup> Edition, TMH Education.

**Pre Lab/ Prior Concepts:**

### **Convolution**

Discrete time convolution is a method of finding response of linear time invariant system. It is based on the concepts of linearity and time invariance and assumes that the system information

is known in terms of its impulse response  $h[n]$ .

Convolution is defined as

$$Y[n] = \sum_{k=-\infty}^{\infty} h[k]x[n-k] = h[n] * x[n]$$

Convolution consists of folding, shifting, Multiplication and summation operations.

### Circular Convolution

Circular convolution between two length N sequences can be carried out as shown by the expression below:

$$y_c[n] = \sum_{m=0}^{N-1} g[m]h[\langle n-m \rangle_N]$$

Since the above operation involves two length-N sequences it is referred to as the N-point circular convolution and denoted by:

$$y_c[n] = g[n] \circledast_N h[n]$$

As in linear convolution circular convolution is commutative.

i.e.

$$g[n] \circledast_N h[n] \equiv h[n] \circledast_N g[n]$$

## Implementation Of Linear Convolution:

```
x = input('Enter x(n) as a vector: ');
h = input('Enter h(n) as a vector: ');

Nx = length(x);
Nh = length(h);
Ny = Nx + Nh - 1;

y = zeros(1, Ny);

for n = 0:Ny-1
    sum_val = 0;
    for k = 0:Nx-1
        if (n-k >= 0) && (n-k < Nh)
            sum_val = sum_val + x(k+1) * h(n-k+1);
        end
    end
    y(n+1) = sum_val;
end

y_conv = conv(x, h);

nx = 0:Nx-1;
nh = 0:Nh-1;
ny = 0:Ny-1;

figure;

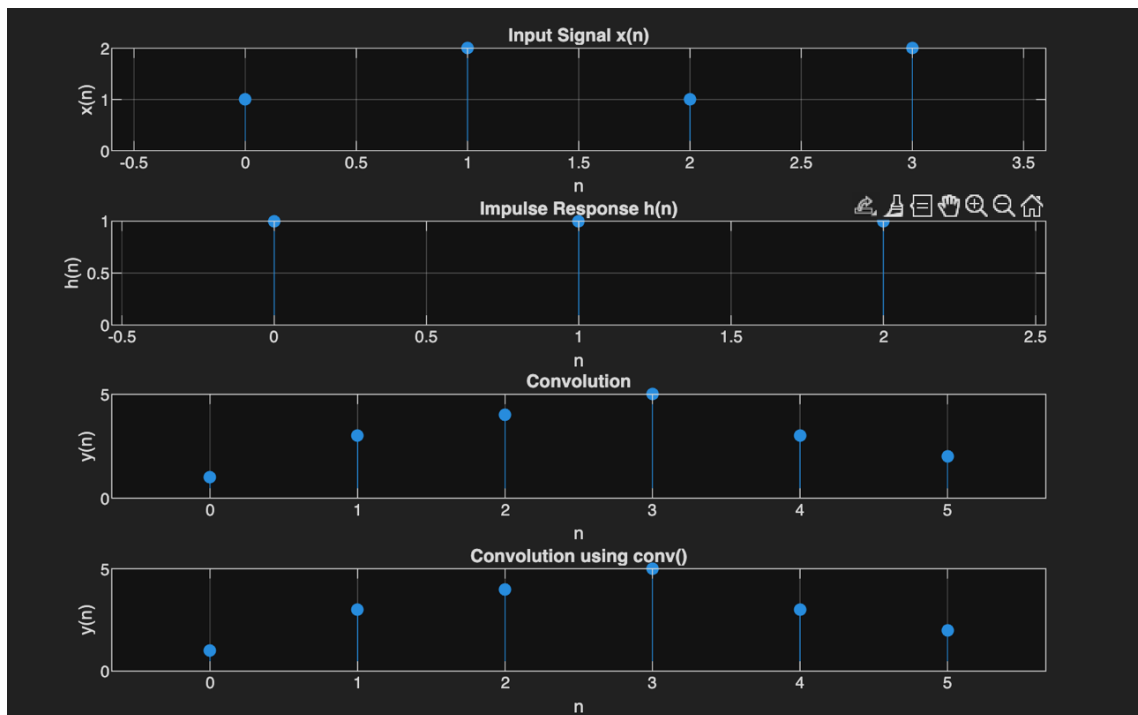
subplot(4,1,1)
stem(nx, x, 'filled')
title('Input Signal x(n)')
xlabel('n')
ylabel('x(n)')
grid on

subplot(4,1,2)
stem(nh, h, 'filled')
title('Impulse Response h(n)')
xlabel('n')
ylabel('h(n)')
grid on

subplot(4,1,3)
stem(ny, y, 'filled')
title('Convolution')
xlabel('n')
ylabel('y(n)')
grid on

subplot(4,1,4)
stem(ny, y_conv, 'filled')
title('Convolution using conv()')
xlabel('n')
ylabel('y(n)')
grid on
```

## Output-



```

Enter x(n) as a vector: [1 2 1 2]
Enter h(n) as a vector: [1 1 1]

y_conv =

     1     3     4     5     3     2

>>
  
```

### Example Of Linear Convolution:

Linear Convolution

$$x(n) = [1 \ 2 \ 1 \ 2]$$

$$h(n) = [1 \ 1 \ 1]$$

$$L - M + 1 = 4 + 3 - 1 = 6$$

$$\Rightarrow y = 0 \text{ to } 5$$
  

$$x(k) = [1 \ 2 \ 1 \ 2]$$

$$h(k) = [1 \ 1 \ 1]$$
  

$$y(n) = \sum_{k=-\infty}^{\infty} x(k) * h(n-k)$$
  

$$y(0) = \sum_{k=-\infty}^{\infty} x(k) * h(0-k)$$
  

$$h(0-k) \Rightarrow [1 \ 1 \ 1]$$

$$\therefore x(k)h(0-k) \Rightarrow [1 \ 1 \ 1]$$
  

$$\boxed{y(0) = 1}$$
  

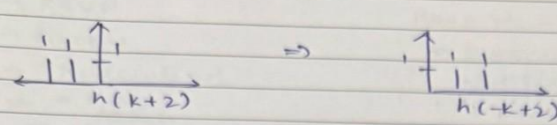
$$y(1) = \sum_{k=-\infty}^{\infty} x(k) * h(1-k)$$
  

$$h(-k+1) \Rightarrow [1 \ 1 \ 1] \Rightarrow [1 \ 1 \ 1]$$
  

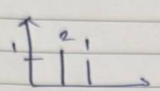
$$x(k)h(1-k) \Rightarrow [1 \ 2 \ 1]$$
  

$$\therefore \boxed{y(1) = 1 + 2 = 3}$$

$y(2) = \sum_{k=-\infty}^{\infty} x(k) h(2-k)$

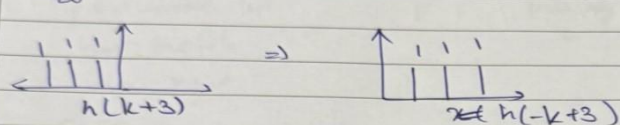


$x(k) h(2-k)$

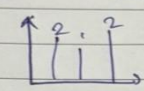


$y(2) = 1+2+1=4$

$y(3) = \sum_{k=-\infty}^{\infty} x(k) h(3-k)$

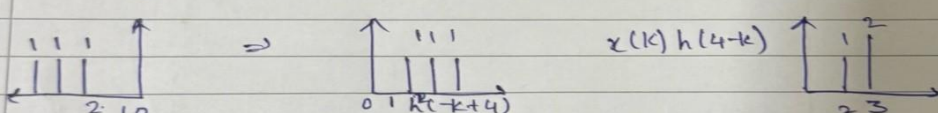


$x(k) h(3-k) =$

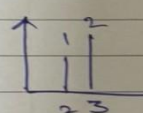


$y(3) = 2+2+1=5$

$y(4) = \sum_{k=-\infty}^{\infty} x(k) h(4-k)$

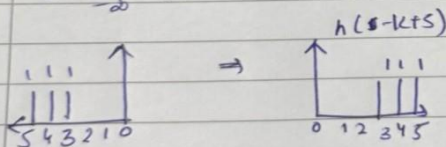


$x(k) h(4-k)$

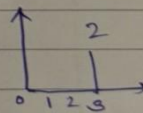


$y(4) = 1+2+2=5$

$y(5) = \sum_{k=-\infty}^{\infty} x(k) h(5-k)$



$x(k) h(5-k)$



$y(5) = 2$

$y(n) = [1, 3, 4, 5, 3, 2]$

## Implementation Of Circular Convolution:

```
x = input('Enter x(n) as a vector: ');
h = input('Enter h(n) as a vector: ');

N=length(x);

H = zeros(N, N);
for i = 1:N
    H(:, i) = circshift(h.', i-1);
end

disp(H)

y = x * H;
y = y.';

y_circ = cconv(x, h, N);

disp(y_circ)

n = 0:N-1;

figure;

subplot(4,1,1)
stem(n, x, 'filled')
title('Input Signal x(n)')
xlabel('n')
ylabel('x(n)')
grid on

subplot(4,1,2)
stem(n, h, 'filled')
title('Input Signal h(n)')
xlabel('n')
ylabel('h(n)')
grid on

subplot(4,1,3)
stem(n, y, 'filled')
title('Circular Convolution')
xlabel('n')
ylabel('y(n)')
grid on

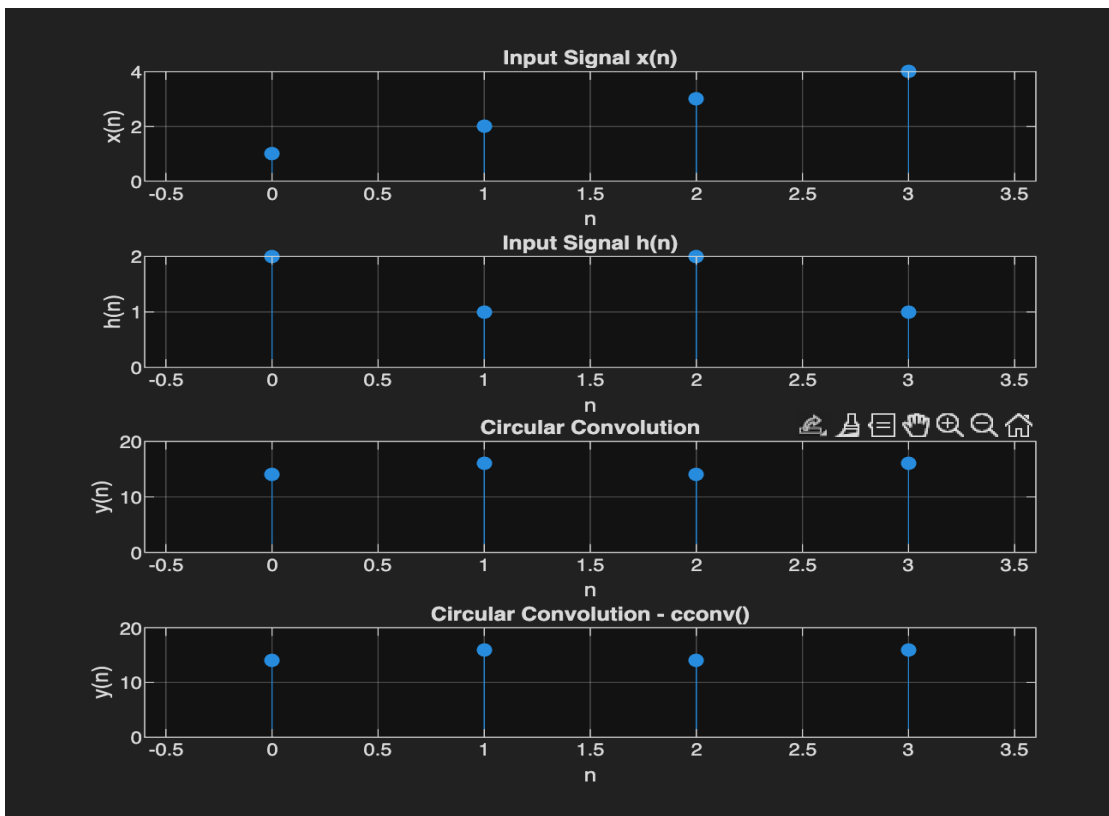
subplot(4,1,4)
stem(n, y_circ, 'filled')
title('Circular Convolution - cconv()')
xlabel('n')
ylabel('y(n)')
grid on
```

## Output-

```
Enter x(n) as a vector: [1 2 3 4]
Warning: Function plot has the same name as a MATLAB built-in. We suggest you rename the function to avoid a potential name conflict.
> In circconvo (line 4)
Enter h(n) as a vector: [2 1 2 1]
 2     1     2     1
 1     2     1     2
 2     1     2     1
 1     2     1     2

 14    16    14    16

>>
```



### Example Of Circular Convolution:

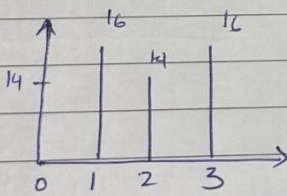
Circular Convolution

$$x(n) = [1 \ 2 \ 3 \ 4]$$

$$h(n) = [2 \ 1 \ 2 \ 1]$$

$$H(n) = \begin{matrix} & 0 & 1 & 2 & 3 \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 2 & 1 & 2 & 1 \\ 1 & 2 & 1 & 2 \\ 2 & 1 & 2 & 1 \\ 1 & 2 & 1 & 2 \end{bmatrix} \end{matrix}$$

$$H(n) \cdot x(n) = \begin{bmatrix} 2 & 1 & 2 & 1 \\ 1 & 2 & 1 & 2 \\ 2 & 1 & 2 & 1 \\ 1 & 2 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} =$$

$$\begin{bmatrix} 2+2+6+4 \\ 1+4+3+8 \\ 2+2+6+4 \\ 1+4+3+8 \end{bmatrix} = \begin{bmatrix} 14 \\ 16 \\ 14 \\ 16 \end{bmatrix}$$


$$y(n) = [14 \ 16 \ 14 \ 16]$$

**Conclusion:-**

In this experiment, linear and circular convolution of discrete-time signals were successfully implemented using MATLAB. The graphical representation of input signals and their convolution outputs aided in visual comparison and better conceptual clarity. Thus, the experiment helped in understanding the fundamental concept of convolution and its practical implementation using MATLAB.

**Date:** \_\_\_\_\_

**Signature of faculty in-charge**

### Post Lab Descriptive Questions

1. Explain the role of convolution in signal processing.

Convolution is a core concept in signal processing because it helps find the output of a linear time-invariant (LTI) system when the input signal and the system's impulse response are known. It explains how one signal influences and reshapes another. Convolution is commonly used in areas like filtering, system analysis, image processing, audio processing, and communication systems. It is useful for studying how signals change and how systems behave in both the time domain and the frequency domain.

2. Explain the difference between linear and circular convolution?

| Linear Convolution                               | Circular Convolution                                                      |
|--------------------------------------------------|---------------------------------------------------------------------------|
| It is used to find the response of an LTI system | It is mainly used in digital signal processing and FFT-based computations |
| Output length is ( $L + M - 1$ )                 | Output length is ( $N$ )                                                  |
| No periodicity is assumed                        | Signals are assumed to be periodic                                        |
| Computed directly using convolution sum          | Computed using modulo indexing                                            |
| Used in time-domain system analysis              | Used in frequency-domain processing                                       |

3. Explain with the help of an example the steps required to transform linear convolution with circular convolution and vice-versa.

To compute **linear convolution using circular convolution**, both input sequences are first **zero-padded** so that their length becomes at least  $L + M - 1$  (where  $L$  and  $M$  are the lengths of the two sequences). Then, **circular convolution** is carried out using this padded length, which produces the **same result as linear convolution**.

On the other hand, **circular convolution can be derived from linear convolution by folding (wrapping)** the linear convolution output into a chosen fixed length. This wrapping creates a **periodic effect**, which transforms the linear result into a circular one.

#### Example:

If two sequences of lengths **4** and **3** are linearly convolved, the output length becomes **6**. If we choose  $N = 4$  and wrap the extra samples **modulo-4**, we obtain the **circular convolution** result.

